

Purple Team Detection Engineering Lab — Engagement Report

Author: Jose Angel Rodriguez **Engagement type:** Self-directed purple team exercise & detection engineering **Environment:** Isolated virtual lab (lab.local Active Directory domain) **Date:** June 2026 **Classification:** Portfolio / Demonstration

1. Executive Summary

This report documents the design, execution, and outcomes of a self-built **purple team detection engineering lab**. The objective was to stand up a realistic Active Directory enterprise, emulate adversary behavior against it, engineer detections from the resulting telemetry, and deliver real-time alerting to a SOC notification channel — exercising the full security-operations lifecycle from data collection to response.

Working from both the offensive and defensive sides, I executed **reconnaissance, enumeration, and execution techniques** mapped to MITRE ATT&CK, then authored **five Splunk detections** that successfully identified that activity. Detections were operationalized as scheduled alerts that push to a **Discord SOC channel** via webhook. The lab demonstrably detected a live attacker — capturing the Kali host’s source IP and the account used in a failed-logon attempt — and generated **122 confirmed alert firings** for the PowerShell execution detection alone.

The exercise also surfaced and resolved several real infrastructure problems (host-only firewall profile mismatch, case-sensitive index routing, Windows 11 deployment constraints), each of which is documented in Section 9 as evidence of hands-on troubleshooting.

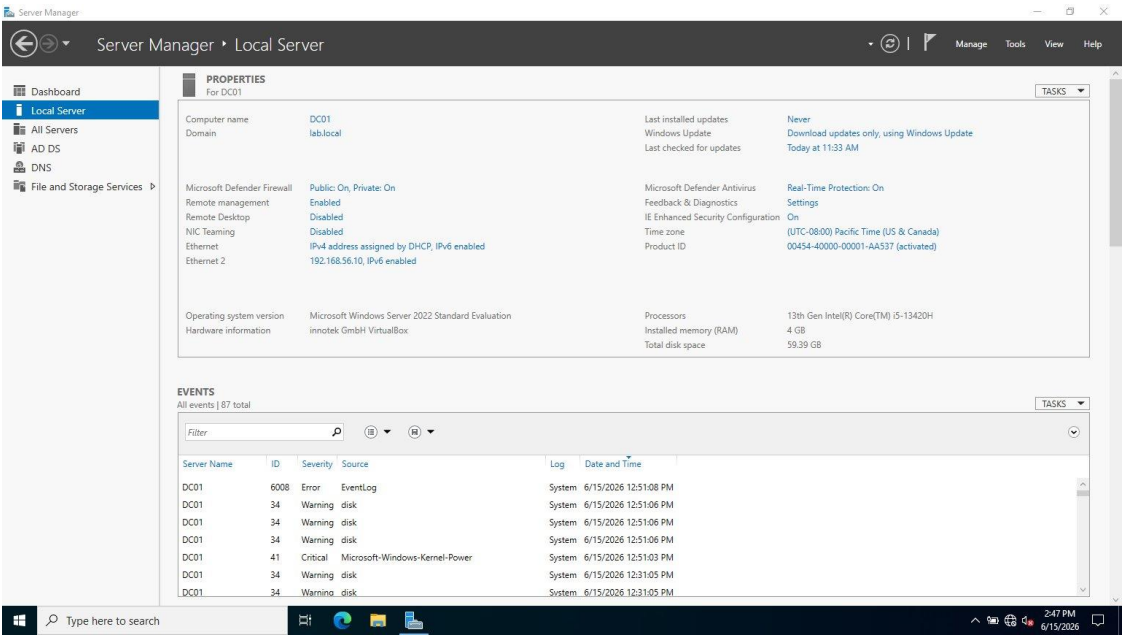
2. Environment & Architecture

The lab was built in Oracle VirtualBox on a single host (13th-gen Intel i5), using an isolated **host-only network (192.168.56.0/24)** for lab traffic plus NAT adapters for controlled internet access. The host machine runs the Splunk indexer, reachable to the VMs at 192.168.56.1.

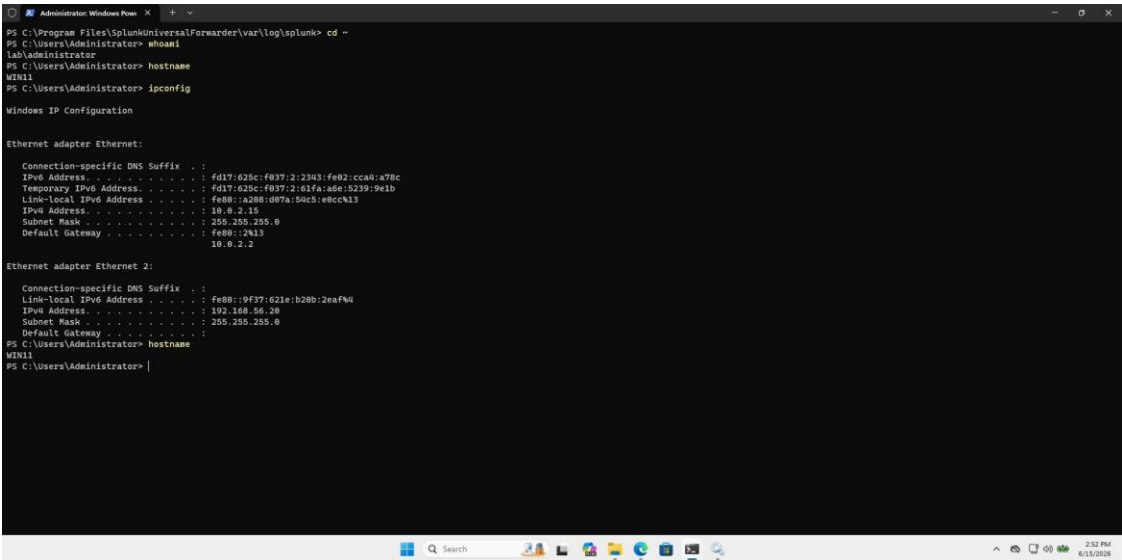
Host	Role	Operating System	Lab IP	Key Software
DC01	Domain Controller / DNS	Windows Server 2022 Standard	192.168.56.10	AD DS, DNS, lab.local
WIN11	Endpoint (victim)	Windows 11 Pro	192.168.56.20	Sysmon, Splunk Universal Forwarder
Splunk	SIEM / Indexer	Host OS	192.168.56.1	Splunk Enterprise

Host	Role	Operating System	Lab IP	Key Software
NAPLS589	Attacker	Kali Linux	192.168.56.103	nmap, enum4linux, smbclient

Domain: lab.local (NetBIOS LAB). Organizational Units were created for **HR, IT, and Finance**, populated with realistic user accounts (e.g., rgonzalez, plus a privileged IT account) to provide authentication activity and credential targets.



DC01 — Windows Server 2022 domain controller (lab.local), static IP 192.168.56.10



WIN11 endpoint — dual-adapter networking and domain user (lab\administrator)

```
Session Actions Edit View Help
nameserver 8.8.4.4

[naples@990 NAPLES99]~$ cat /etc/resolv.conf
# Generated by NetworkManager
nameserver 8.8.8.8
nameserver 8.8.4.4

[naples@990 NAPLES99]~$ ping -c 3 192.168.56.10
ping: invalid argument: '192.168.56.10'

[naples@990 NAPLES99]~$ ping -c 4 192.168.56.10
PING 192.168.56.10 (192.168.56.10) 56(84) bytes of data:
64 bytes from 192.168.56.10: icmp_seq=1 ttl=120 time=3.58 ms
64 bytes from 192.168.56.10: icmp_seq=2 ttl=120 time=0.927 ms
64 bytes from 192.168.56.10: icmp_seq=3 ttl=120 time=1.14 ms
64 bytes from 192.168.56.10: icmp_seq=4 ttl=120 time=0.535 ms

--- 192.168.56.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3019ms
rtt min/avg/max/mdev = 0.535/2.846/5.582/2.852 ms

[naples@990 NAPLES99]~$ no-connection-editor
(no-connection-editor:4728): dconf-WARNING **: 16:43:00.330: unable to open file '/etc/dconf/db/local': failed to open file "/etc/dconf/db/local": open() failed: No such file or directory; expect degraded performance

[naples@990 NAPLES99]~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:1f:58:10 brd ff:ff:ff:ff:ff:ff
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:1f:58:10 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.10/24 brd 192.168.56.255 scope global dynamic noprefixroute eth1
        valid_lft 473sec preferred_lft 473sec
    inet6 fe80::a0b:27ff:fe58:10/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

[naples@990 NAPLES99]~$ ip netns exec lab.local 192.168.56.10
Server: 192.168.56.10
Address: 192.168.56.10853

Name: lab.local
Address: 192.168.56.10
Name: lab.local
Address: 10.0.2.15
Name: lab.local
Address: fd17:625c:f037:2:96ab:50a7:140a:8309

[naples@990 NAPLES99]~$
```

Kali attacker — host-only connectivity and DNS resolution against the DC

3. Telemetry & Data Pipeline

Visibility is the foundation of the lab. The endpoint was instrumented and its logs forwarded to Splunk for indexing and search.

Collection (WIN11 endpoint):

- **Sysmon** (SwiftOnSecurity configuration) — process creation, network connections, image loads, registry activity, and more.
- **Windows Event Logs** — Security, System, and PowerShell/Operational channels.

Transport & indexing:

- **Splunk Universal Forwarder** ships events to the indexer at 192.168.56.1:9997.
- Events are routed by inputs.conf into dedicated indexes: **sysmon** and **windows**.
- The **Splunk Add-on for Microsoft Windows** and **Splunk Add-on for Sysmon** provide CIM-compliant field extraction.

At steady state the endpoint produced thousands of process-creation events (Sysmon Event ID 1), confirming a healthy, high-fidelity pipeline suitable for detection development.

Time	Event
6/15/2026 4:43:59.162 PM	LogName:Microsoft-Windows-Sysmon/Operational
4:43:59.162 PM	EventCode:1
	EventID:4
	ComputerName:WIN11.lab.local
	User:NOT_TRANSLATED
	Sid:S-1-5-18
	SidType:0
	SourceName:Microsoft-Windows-Sysmon
	Type:Information
	RecordNumber:8649
	Keywords:None
	TaskCategory:Sysmon config state changed
	OpCode:Info
	Message:Process Create:
	ProcessName:
	UtcTime: 2025-06-15 22:43:59.159
	ProcessGuid: {F862ea90-802f-6a30-b40f-000000000000}
	ProcessId: 5892
	Image: C:\Program Files\SplunkUniversalForwarder\bin\splunk-regmon.exe
	FileVersion: 10.4.0
	Description: Registry monitor
	Product: splunk application
	Company: Splunk Inc.
	OriginalFileName: splunk-regmon.exe
	CommandLine: "C:\Program Files\SplunkUniversalForwarder\bin\splunk-regmon.exe"
	CurrentDirectory: C:\WINDOWS\system32\
	User: NT AUTHORITY\SYSTEM
	LogonGuid: {F862ea90-5718-6a30-e783-000000000000}
	LogonId: 0x3c7
	TerminalSessionId: 0
	IntegrityLevel: System
	Hashes: MD5-BEE6270B59426EBEE6117BF895650678_SHA256-NEC5822948CE9BF74CCT8159898868020840F76151F85DA86321A5F39CAB_IMPHASH-404F60C3F27FC
	ParentProcessGuid: {F862ea90-698a-6a30-148a-000000000000}
	ParentProcessId: 7744
	ParentImage: C:\Program Files\SplunkUniversalForwarder\bin\splunkd.exe

Raw Sysmon Event ID 1 (process creation) indexed in Splunk from WIN11.lab.local

4. Red Team Operations

All offensive activity was conducted from the Kali attacker (192.168.56.103) against the lab, or via adversary-emulation tooling run locally on the endpoint. Each action is mapped to MITRE ATT&CK.

4.1 Network Service Discovery — T1046

An nmap service/OS scan (-sV -o -sS) was run against the endpoint, identifying exposed services (MSRPC on 135, plus 5040 and 7680) and fingerprinting the host as Windows 11.

```

Session Actions Edit View Help

[nmap5890@NAPL5890:~/Documents/Hacking/lab.local]
$ enumlinux -s $IP
Starting enumlinux v0.9.1 ( http://labs.portcullis.co.uk/application/enumlinux/ ) on Mon Jun 15 20:56:04 2026

----- ( Target Information ) -----
Target ..... 192.168.56.20
RID Range ..... 500-550,1000-1050
Username ..... ''
Password ..... ''
Known Usernames .. administrator, guest, kbrigt, domain admins, root, bin, none

----- ( Enumerating Workgroup/Domain on 192.168.56.20 ) -----

----- ( Nbtstat Information for 192.168.56.20 ) -----

Looking up status of 192.168.56.20
No reply from 192.168.56.20

----- ( Session Check on 192.168.56.20 ) -----

[E] Server doesn't allow session using username '', password ''. Aborting remainder of tests.

[nmap5890@NAPL5890:~/Documents/Hacking/lab.local]
$ sudo cd ~/el

```

```

[nmap5890@NAPL5890:~/Documents/Hacking/lab.local]
$ cd ~/Documents/Hacking/lab.local
$ cat nmap.nmap
cat: nmap.nmap: No such file or directory

[nmap5890@NAPL5890:~/Documents/Hacking/lab.local]
$ ls
nmap

[nmap5890@NAPL5890:~/Documents/Hacking/lab.local]
$ cat nmap
# Nmap 7.95 scan initiated Mon Jun 15 20:55:58 2026 as: /usr/lib/nmap/nmap --privileged -Pn -sV -o -sS -p- -T4 -uN n
map -v 192.168.56.20
Map scan report for 192.168.56.20
Host is up (9.003s latency).
Not shown: 65532 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp    open  msrpc
5040/tcp   open  unknown
7680/tcp   open  pando-pub?
Mac Address: 08:00:27:09:70:7F (NCS Systemtechnik/Oracle VirtualBox virtual NIC)
Warning: OSscan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: general purpose
Running (OS fingerprinting): Microsoft Windows 11 (Q13), FreeBSD 6.X (80X)
OS CPE: cpe:/o:microsoft:windows_11 cpe:/o:freebsd:freebsd:6.2
Aggressive OS guesses: Microsoft Windows 11 21H2 (Q13), FreeBSD 6.2-RELEASE (80X)
No exact OS matches for host (test conditions non-ideal).
Uptime guess: 0.284 days (since Mon Jun 15 13:49:10 2026)
Network Distance: 3 hops
TCP Sequence Prediction: Difficulty=263 (Good luck!)
IP ID Sequence Generation: Incremental
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Read data files from: /usr/share/nmap
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
# Nmap done at Mon Jun 15 20:58:43 2026 -- 1 IP address (1 host up) scanned in 363.27 seconds

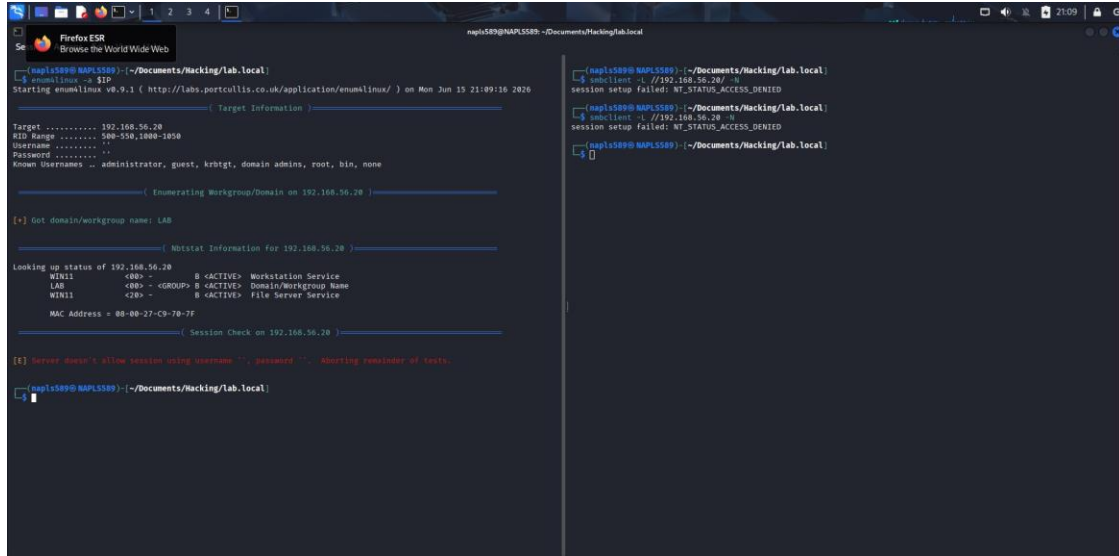
[nmap5890@NAPL5890:~/Documents/Hacking/lab.local]
$

```

Network Service Discovery (T1046) — nmap scan of the endpoint from Kali

4.2 Account & Domain Discovery — T1087 / T1482

enum4linux -a was run against the endpoint, successfully recovering the **domain/workgroup name (LAB)**, NetBIOS service records (Workstation Service, File Server Service), and the host MAC address.



```
(nmap1589@NAPL5589) [~/Documents/Hacking/lab.local]
$ enum4linux -a $IP
Starting enum4linux v0.9.1 ( http://labb.portcullis.co.uk/application/enum4linux/ ) on Mon Jun 15 21:09:16 2020

( Target Information )
Target ..... 192.168.56.20
RID Range ..... 500-550,1000-2050
Username ..... ''
Password ..... ''
Known Users ..... administrator, guest, krbtgt, domain admins, root, bin, none

( Enumerating Workgroup/Domain on 192.168.56.20 )

[+] Get domain/workgroup name: LAB

( Netstat Information for 192.168.56.20 )

Looking up status of 192.168.56.20
WIN11 <00> B <ACTIVE> Workstation Service
LAB <00> B <ACTIVE> Domain/Workgroup Name
WIN11 <20> B <ACTIVE> File Server Service
MAC Address = 88-00-27-C9-70-7F

( Session Check on 192.168.56.20 )

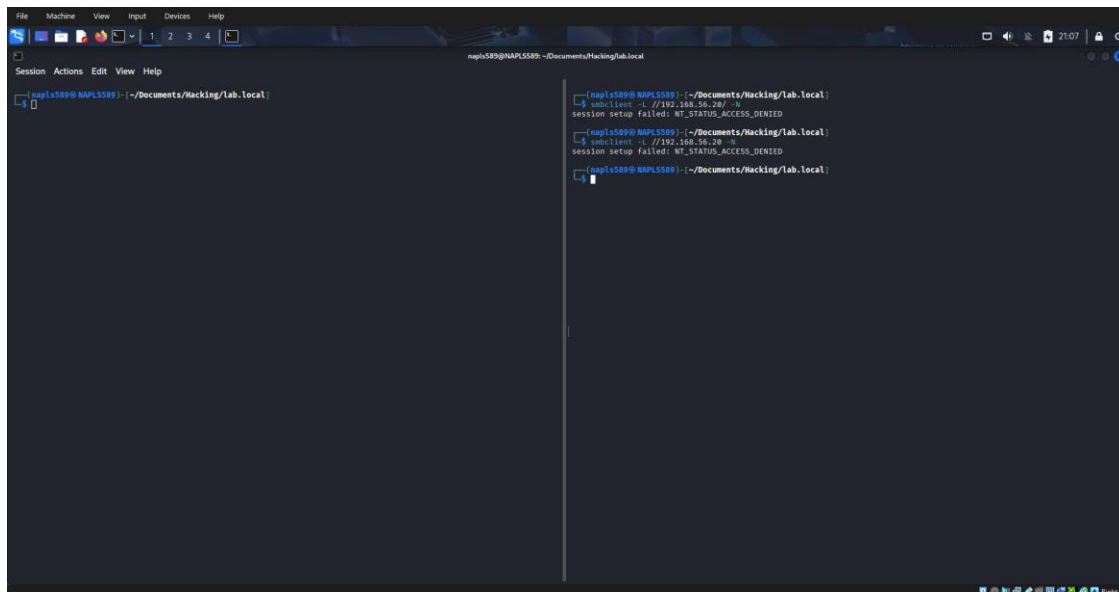
[!] Server doesn't allow session using username '', password ''. Aborting remainder of tests.

(nmap1589@NAPL5589) [~/Documents/Hacking/lab.local]
$ smbclient -L //192.168.56.20 -N
session setup failed: NT_STATUS_ACCESS_DENIED
(nmap1589@NAPL5589) [~/Documents/Hacking/lab.local]
$ smbclient -L //192.168.56.20 -N
session setup failed: NT_STATUS_ACCESS_DENIED
(nmap1589@NAPL5589) [~/Documents/Hacking/lab.local]
$
```

Account/Domain Discovery (T1087) — enum4linux recovering the LAB domain

4.3 Network Share Discovery — T1135

SMB share enumeration was attempted with `smbclient -L //192.168.56.20 -N`. The endpoint correctly rejected the anonymous session (NT_STATUS_ACCESS_DENIED) — and, importantly, the *attempt itself* generated the telemetry that triggered detection.

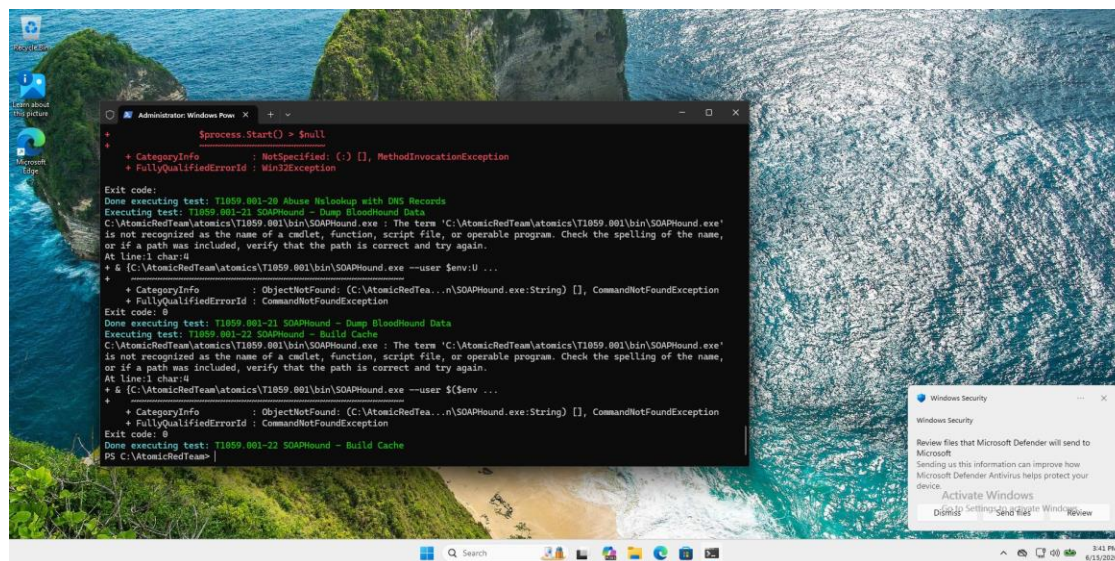


```
(nmap1589@NAPL5589) [~/Documents/Hacking/lab.local]
$ smbclient -L //192.168.56.20 -N
session setup failed: NT_STATUS_ACCESS_DENIED
(nmap1589@NAPL5589) [~/Documents/Hacking/lab.local]
$ smbclient -L //192.168.56.20 -N
session setup failed: NT_STATUS_ACCESS_DENIED
(nmap1589@NAPL5589) [~/Documents/Hacking/lab.local]
$
```

Network Share Discovery (T1135) — smbclient SMB enumeration attempts

4.4 Command & Scripting Interpreter: PowerShell — T1059.001

Atomic Red Team (Invoke-AtomicRedTeam v2.3.0) was installed on the endpoint and used to execute a battery of T1059.001 atomic tests, including SOAPHound/SharpHound BloodHound data collection, NTFS Alternate Data Stream execution, New-PSSession, encoded-command variations, and Nslookup-over-DNS abuse. These produced rich, adversary-representative PowerShell command lines under the LAB\Administrator context.



Adversary emulation — Atomic Red Team T1059.001 PowerShell tests executing

4.5 Credential Access via Failed Authentication — T1110 / T1078

Authentication failures were generated against the environment, including attempts from the Kali host. These surfaced as Windows Event ID **4625** records carrying the attacker's source IP and the account names used.

5. Detection Engineering

Five detections were authored as Splunk saved searches under a consistent **PurpleTeam_** naming convention, each scheduled to run on a cron interval and configured to alert when results are returned. All five are enabled.

Name	Actions	Type	Next Scheduled Time	Display View	Owner	App	Alerts	Sharing	Status
PurpleTeam_AtomicRedTeam_Event-Detection	Edit Run View Record	Alert	2025-06-15 19:23:00 Central America Standard Time	none	jrod589	search	122	App	✓ Enabled
PurpleTeam_Failed_Logon_Detection	Edit Run View Record	Alert	2025-06-15 19:23:00 Central America Standard Time	none	jrod589	search	1	Private	✓ Enabled
PurpleTeam_Network_Logon_Detection	Edit Run View Record	Alert	2025-06-15 19:23:00 Central America Standard Time	none	jrod589	search	0	Private	✓ Enabled
PurpleTeam_Network_Recon_Detection	Edit Run View Record	Alert	none	none	jrod589	search	0	Private	✓ Enabled
PurpleTeam_Share_Enumeration_Detection	Edit Run View Record	Alert	2025-06-15 19:23:00 Central America Standard Time	none	jrod589	search	0	Private	✓ Enabled

Detection inventory — five enabled PurpleTeam_Splunk alerts

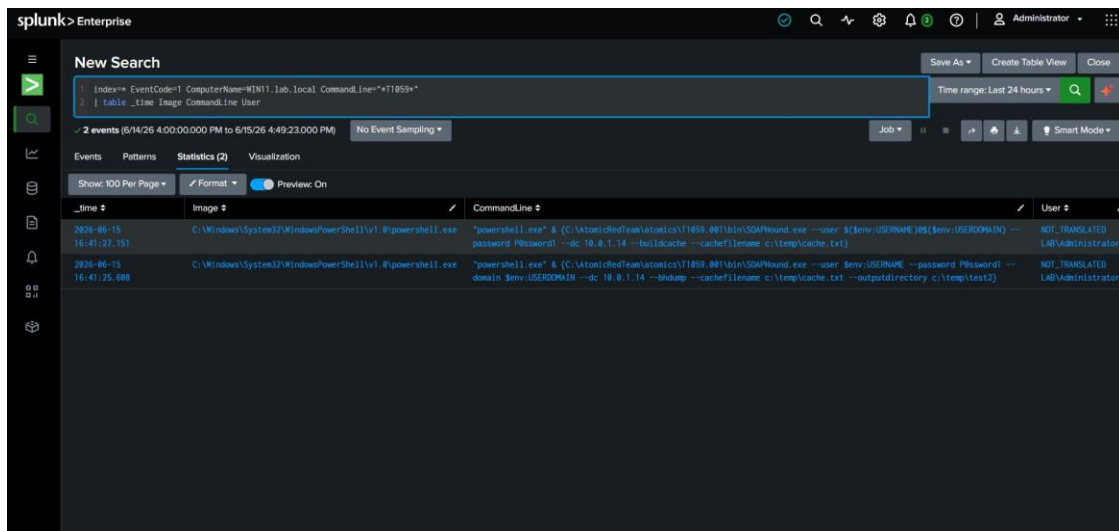
Detection	ATT&CK	Trigger logic (summary)	Primary data source
PurpleTeam_AtomicRedTeam_Event-Detection	T1059.001	PowerShell/MSHTA execution patterns from Atomic Red Team activity	Sysmon EID 1
PurpleTeam_Failed_Logon_Detection	T1110 / T1078	Failed authentication events	Security EID 4625
PurpleTeam_Network_Logon_Detection	T1021	Network (Type 3) logon activity	Security EID 4624
PurpleTeam_Network_Recon_Detection	T1046	Network/service discovery indicators	Network / Sysmon EID 3
PurpleTeam_Share_Enumeration_Detection	T1135	SMB share enumeration / anonymous session attempts	Security / Sysmon

5.1 PowerShell Execution Detection (T1059.001)

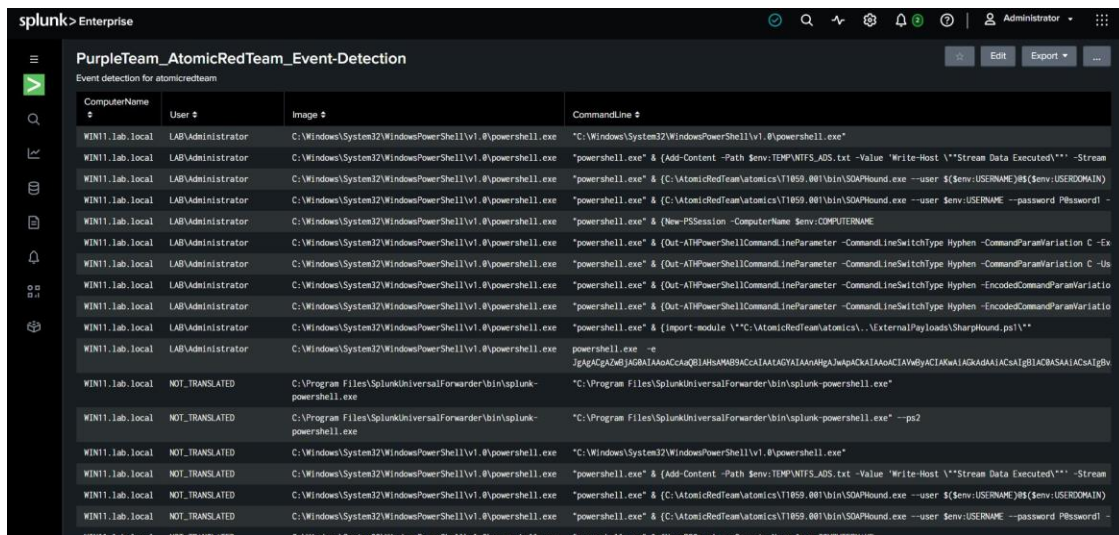
The flagship detection identifies Atomic Red Team’s PowerShell execution by correlating Sysmon process-creation events against suspicious command-line content:

```
index=* EventCode=1 ComputerName=WIN11.lab.local CommandLine="*T1059*"
| table _time Image CommandLine User
```

A dedicated dashboard panel surfaces the matching executions (image, command line, user) for analyst triage. In production runs this detection fired **122 times**, cleanly attributing the activity to LAB\Administrator on WIN11.lab.local.



PowerShell execution detection (T1059.001) correlating Sysmon command lines



Analyst dashboard surfacing detected Atomic Red Team executions

5.2 Failed Logon Detection (T1110) — Live Attacker Capture

This detection proved its value by identifying the live attacker. Querying failed-logon events surfaced authentication failures originating from the **Kali host (192.168.56.103)** under the account **nap1s589**, alongside other failed attempts — directly tying observed malicious traffic to a source IP and account:

```
index=* EventCode=4625
| table _time Source_Network_Address Account_Name Failure_Reason
| sort -_time
```


New Search

```
1 | index=* EventCode=4625
2 | table _time Source_Network_Address Account_Name Failure_Reason
3 | sort -_time
```

7 events (6/14/26 8:00:00.000 PM to 6/15/26 8:11:00.000 PM) No Event Sampling

Events Patterns Statistics (7) Visualization

Show: 100 Per Page Format Preview: On

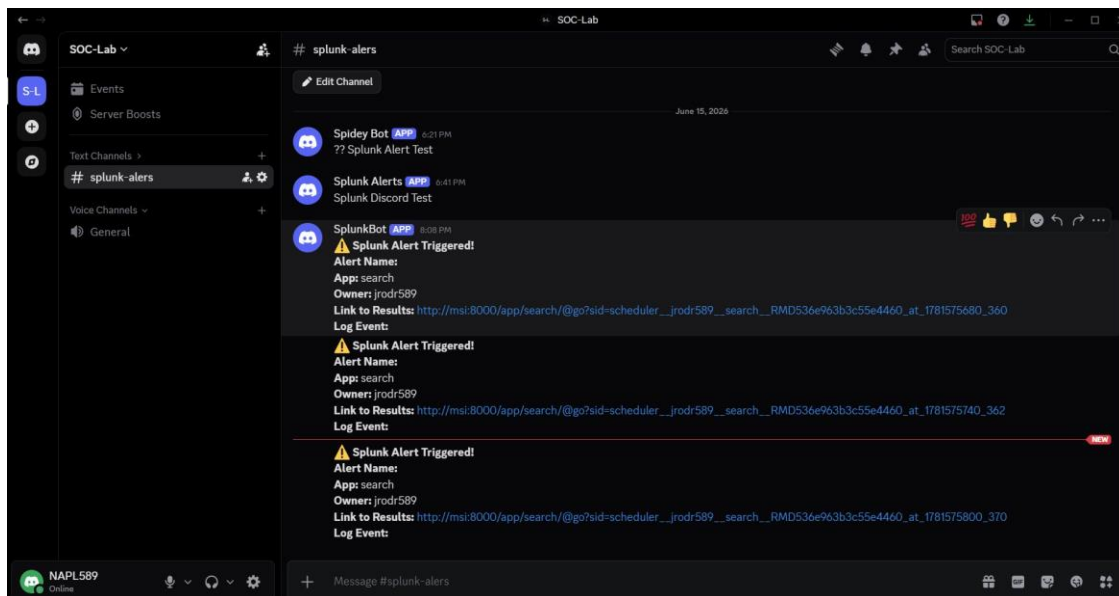
_time	Source_Network_Address	Account_Name	Failure_Reason
2025-06-15 20:07:48.499	192.168.56.103	nagisa89	Unknown user name or bad password.
2025-06-15 20:07:35.143	192.168.56.103	-	Unknown user name or bad password.
2025-06-15 12:58:32.363	127.0.0.1	WIN11\$ lab_admin	An Error occurred during Logon.
2025-06-14 22:47:54.854	127.0.0.1	WIN11\$ rgonzalez@lab.local	Unknown user name or bad password.
2025-06-14 22:46:05.842	127.0.0.1	WIN11\$ rgonzalez@lab.local	Unknown user name or bad password.
2025-06-14 22:45:58.282	127.0.0.1	WIN11\$ rgonzalez@lab.local	Unknown user name or bad password.
2025-06-14 22:45:05.445	127.0.0.1	WIN11\$ rgonzalez	Unknown user name or bad password.

Live attacker capture — failed-logon detection (4625) recovering the Kali source IP and account

6. Alerting & Automated Response (Splunk → Discord)

Each detection’s scheduled-search firing invokes a **webhook alert action** that posts to a dedicated **Discord SOC channel** (#splunk-alers in the *SOC-Lab* server). Messages are delivered by a Splunk webhook integration and render as structured “**Splunk Alert Triggered!**” notifications containing the alert name, app, owner, and a deep link back to the originating search results in Splunk.

This closes the loop from **telemetry** → **detection** → **real-time analyst notification**, mirroring how a production SOC routes alerts into a chat-ops channel for triage.



Automated alerting — Splunk detection firing into the Discord SOC channel via webhook

7. MITRE ATT&CK Coverage

Tactic	Technique	ID	Emulated	Detected
Discovery	Network Service Discovery	T1046	✓	✓
Discovery	Account/Domain Discovery	T1087 / T1482	✓	✓
Discovery	Network Share Discovery	T1135	✓	✓
Execution	PowerShell	T1059.001	✓	✓
Credential Access	Brute Force / Valid Accounts	T1110 / T1078	✓	✓
Lateral Movement	Remote Services	T1021	✓	✓

8. Results & Metrics

- **5** original detections authored, scheduled, and enabled.
- **122** confirmed firings of the PowerShell execution detection.
- **Live attacker attribution:** captured Kali source IP (192.168.56.103) and account (nap1s589) from failed-logon telemetry.
- **End-to-end automation:** detections deliver real-time alerts to a Discord SOC channel via webhook.
- **Full lifecycle demonstrated:** instrumentation → forwarding → indexing → detection → alerting → analyst notification.

9. Engineering Challenges Overcome

Authentic problem-solving was a core part of this build. Notable issues diagnosed and resolved:

- **Forwarder traffic blocked at the host firewall.** The endpoint could reach the DC but not the indexer on :9997. Systematic testing (Test-NetConnection, netstat, profile checks) isolated the cause to the **host-only network being classified under the “Public” firewall profile**, which the original allow rule didn’t cover. Resolved with an all-profiles inbound rule (-Profile Any).
- **Sysmon data silently dropped.** Three of four channels indexed correctly while Sysmon produced nothing. Root cause was **case-sensitive index routing** — inputs.conf specified index = Sysmon while the index was sysmon, so events were dropped at the indexer. Corrected the case and validated with btool.
- **Windows 11 lab deployment.** Worked through VirtualBox unattended-install failures, TPM 2.0 / EFI requirements, and offline local-account creation to produce a clean, domain-joinable endpoint.

These are documented because they reflect the day-to-day reality of standing up detection infrastructure — and the methodical, evidence-driven approach used to resolve each.

10. Conclusion & Future Work

This lab delivers a working, demonstrable purple team capability: a realistic AD environment, a healthy telemetry pipeline, emulated adversary behavior mapped to MITRE ATT&CK, original detections that caught that behavior (including a live attacker), and automated SOC alerting.

Planned extensions: - Convert detections to **Sigma** rules for vendor-neutral, portable logic. - Add an **ATT&CK Navigator** coverage layer to visualize detection breadth. - Instrument the **DC** with Sysmon and expand to lateral-movement and persistence detections. - Introduce a **C2 framework** (e.g., Sliver) with matching detections — the natural bridge toward offensive security.

Appendix A — Detection Queries

The two queries below are exactly as run in the lab. Replace the representative logic for the remaining three detections with the exact SPL from your saved searches (Splunk → *Settings* → *Searches, Reports, and Alerts* → *Edit*).

PowerShell execution (T1059.001):

```
index=* EventCode=1 ComputerName=WIN11.lab.local CommandLine="*T1059*"
| table _time Image CommandLine User
```

Failed logon (T1110):

```
index=* EventCode=4625
| table _time Source_Network_Address Account_Name Failure_Reason
| sort -_time
```

Appendix B — Indicators of Compromise (Lab)

Type	Value	Context
Source IP	192.168.56.103	Kali attacker (recon, enumeration, failed auth)
Account	nap1s589	Account used in failed-logon attempts
Process	powershell.exe w/ *T1059*, SOAPHound.exe, SharpHound.ps1	Atomic Red Team execution
Tooling	nmap, enum4linux, smbclient	Discovery & enumeration